

2048 GAME WITH USER AUTHENTICATION JavaFX Desktop Application

Course Project Submitted by

Name: Thinn Nu San

Student ID: 25040102003

GitHub Repository: <https://github.com/thinnnusan/2048-javafx>

Project Overview

This project is a complete implementation of the popular 2048 puzzle game developed using **Java 17** and **JavaFX 21**. Unlike basic 2048 versions, this application includes a full user management system with registration, login, high score tracking, and the ability to save and load game progress.

The game provides a smooth and interactive experience with color-coded tiles, proper win/loss detection, and an efficient movement system.

Key Features

- User Registration and Login System
- Persistent High Score Tracking per User
- Save Game & Load Game Functionality
- Classic 2048 gameplay with 4-direction movement
- Efficient board rotation technique for movement logic
- Clean and user-friendly JavaFX interface

Technologies Used

- Java 17
- JavaFX 21
- Maven Build Tool
- Object Serialization for data persistence

Implementation Summary

The project follows a clean modular design with five main classes:

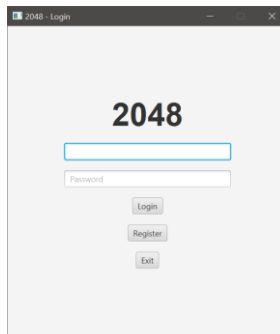
- Main.java – Application entry point
- LoginScreen.java – Handles user authentication
- GameScreen.java – Core game logic and user interface
- User.java – User model with high score
- FileManager.java – Manages saving and loading of users and game data

A notable implementation detail is the **board rotation technique**. Instead of writing four separate movement functions, only `moveLeft()` is implemented. Other directions are achieved by rotating the board, moving left, and rotating back. This approach significantly reduces code duplication and improves maintainability.

The application uses file-based storage (`users.dat` and `savegame.dat`) for persistence.

Example Screenshots

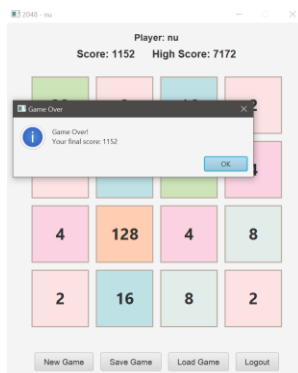
Login Screen



Gameplay Screen



Game Over message



Declaration I hereby declare that this project is my own work and has been developed following academic integrity guidelines.

Signature: _____

Date: 3 June, 2026